

Course Map

Combustion-driven, PVC, adult-supervised

Four lessons and a build, on open ground

A combustion potato launcher is a heat engine that runs once. Fuel and air burn in a closed chamber, the gas gets hot, hot gas pushes hard, and the only thing free to move is the potato. Every lesson here is one term in that sentence, and each one is measured rather than asserted — because the difference between a launcher that works and one that bursts is entirely a matter of numbers you can check beforehand.

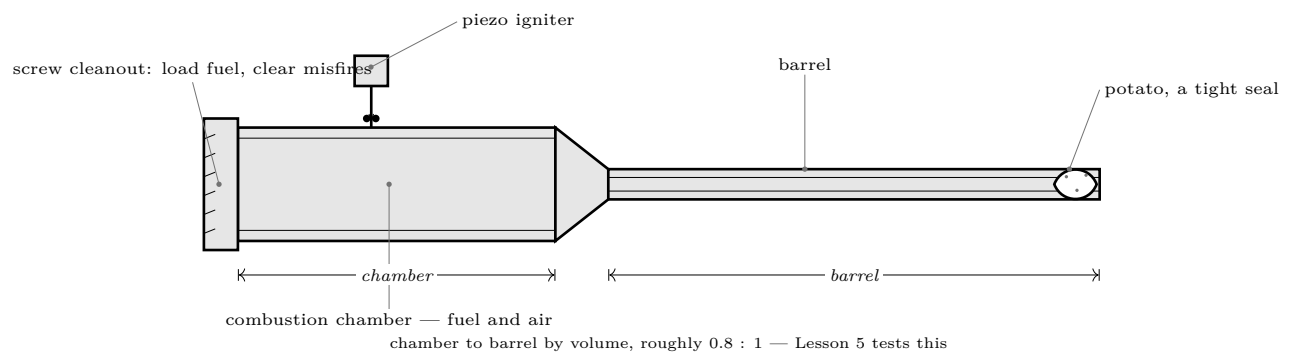
THIS IS A PROJECTILE WEAPON

A combustion launcher throws a half-pound potato several hundred feet at speeds that will kill a person. It is not a toy, it is not a firework, and the fact that it is made of plumbing parts changes nothing about what it does downrange. Every rule on the **Safety** sheet applies every time: adult present and in charge, muzzle discipline as strict as a rifle's, a confirmed clear range, eyes and ears on, and never a person, animal, building, road or anything you cannot afford to destroy in the direction it is pointing.

The through-line

#	Lesson	The idea	Why the launcher needs it
1	Air is stuff	Air has mass, exerts pressure, and pushes in every direction at once. Crush a can with it.	The chamber is full of it, and it is half the fuel.
2	Combustion	Fuel burns only in a narrow band of fuel-to-air ratio. Too little and too much both fail.	This is the whole art of getting a reliable shot.
3	Gas laws	Heat a fixed volume of gas and the pressure rises in proportion. $PV = nRT$.	It says how hard the gas pushes and how much margin the pipe has.
4	Projectile motion	Range, angle, and how to measure a muzzle velocity you cannot see.	It turns a bang into a number, and a number into a safety distance.
5	Build and fire	Assemble, cure, test, and shoot it on open ground.	—

The machine



How each lesson runs

Predict first	Every lesson opens with a question and a box to commit an answer in, before anything is lit. A wrong prediction written down teaches far more than a demonstration nobody doubted.
They build it	The boys cut, deburr, dry-fit, glue and assemble. The adult owns fuel, ignition and the decision to fire.
Measure it	Every lesson produces a number: a pressure, a ratio, a distance, a velocity. Numbers are what separate this from a stunt.
Break it	Change one variable and predict the change. Twice the fuel. Half the barrel. A loose potato.
Then the maths	The equation arrives as the compact way to state what they already measured.

Who does what

Lessons 1–4	Bench work: a can, a candle, a balloon, a tape measure. Nothing here is dangerous beyond ordinary kitchen care.
The build	The boys do essentially all of it. Cutting, deburring, solvent welding and the 48-hour cure are the bulk of the work and none of it involves fuel.
Firing	The adult is in charge of the fuel can, the igniter and the command to fire, permanently. Everyone else is behind the firing line.
Where	Open rural ground with several hundred yards clear downrange, a backstop, and nothing and nobody in the line of fire. This is not a backyard device.
Before anything	Read the Safety sheet start to finish, and check local ordinances — these are legal in most places and restricted in some, and that is on you to confirm before you build.

Pressure-rated pipe only. Schedule 40 *pressure* PVC, printed with an NSF-PW rating and a pressure figure. Never DWV, never foam-core, never cellular-core, never thin-wall drain pipe — those are not rated for internal pressure at all and they fail by shattering into fragments rather than splitting. This is the one specification in the whole project that is not negotiable.

The one sentence version
A measured amount of fuel mixed with the right amount of air burns in a fraction of a second; the gas in the chamber roughly triples in pressure; that pressure acts on the base of the potato all the way down the barrel; and the potato leaves the muzzle carrying the energy the fuel released. Lessons 1 to 4 are the four clauses of that sentence.

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